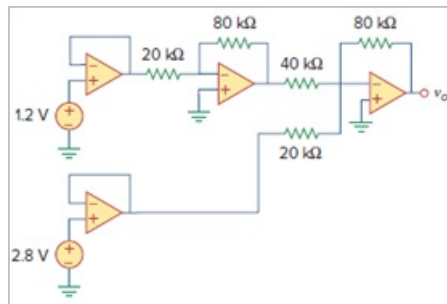


Problem

Obtain the output v_o in the circuit of Fig. 5.94.

Figure 5.94:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.94 in the text book and assign nodes, the modified circuit is shown in Figure 1.

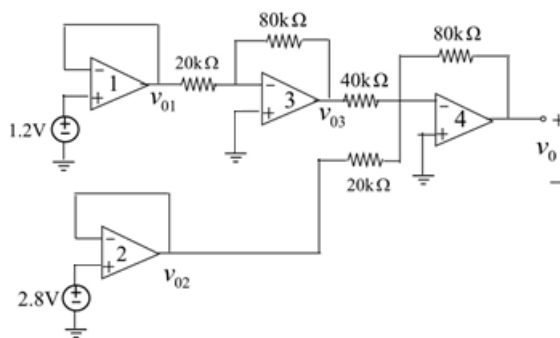


Figure 1

Step 2 of 4

In Figure 1, the 1st and 2nd op-amp circuits are act as an voltage follower circuits. The output voltages are,

$$v_{o1} = 1.2 \text{ V and}$$

$$v_{o2} = 2.8 \text{ V}$$

Step 3 of 4

In Figure 1, the 3rd op-amp circuit is act as an inverting amplifier circuit. The output voltage is,

$$\begin{aligned}v_{o3} &= -\left(\frac{R_f}{R_i}\right)v_i \\&= -\left(\frac{80 \times 10^3}{20 \times 10^3}\right)v_{o1} \\v_{o3} &= -4v_{o1}\end{aligned}$$

Substitute 1.2 V for v_{o1} .

$$\begin{aligned}v_{o3} &= -4(1.2) \\v_{o3} &= -4.8 \text{ V}\end{aligned}$$

Step 4 of 4

In Figure 1, the 4th op-amp circuit is act as a two input summing amplifier circuit. The output voltage is,

$$\begin{aligned}v_o &= -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} \right) \\&= -80 \times 10^3 \left(\frac{v_{o3}}{40 \times 10^3} + \frac{v_{o2}}{20 \times 10^3} \right) \\&= -(2v_{o3} + 4v_{o2})\end{aligned}$$

Substitute -4.8 V for v_{o3} and 2.8 V for v_{o2} .

$$\begin{aligned}v_o &= -(2(-4.8) + 4(2.8)) \\&= -(-9.6 + 11.2) \\&= -(1.6) \\v_o &= -1.6 \text{ V}\end{aligned}$$

Therefore, the output voltage, v_o is -1.6 V .